Neural correlates of word stimulus response in human iEEG

Juee Naik, Michael J. Kahana

Department of Psychology, University of Pennsylvania, Philadelphia, Pennsylvania

Introduction

The word-onset response in free recall

Most intracranial EEG (iEEG) studies of free recall contrast words that are later recalled with words that are forgotten (the subsequent memory effect, SME), or study the retrieval period. As both conditions contain a word, this contrast measures how responses differ with later memory, but cancels out the response to the word itself. Only a few studies have examined the word-onset response directly. Kucewicz et al. (2019) showed that high gamma (HG) activity after word onset moves from visual cortex to the medial temporal lobe (MTL) and then to prefrontal cortex, and Marks et al. (2021) found that theta, alpha/beta and gamma activity each spread across regions on their own timelines during word encoding.

High gamma and alpha as markers of cortical engagement

Broadband HG (70–150 Hz) is widely used as a measure of local population spiking (Manning et al., 2009; Ray and Maunsell, 2011) and is better described as a broadband signal than as an oscillation (Burke et al., 2015). At stimulus onset, alpha and beta power decrease (event-related desynchronization; Pfurtscheller and Lopes da Silva, 1999), which is thought to release cortex from alpha-mediated inhibition (Jensen and Mazaheri, 2010). An alpha/beta decrease together with an HG increase is the typical cortical activation response, first described in sensorimotor cortex (Crone, 1998a,b) and observed in visual cortex (Lachaux et al., 2005), including for pseudowords and consonant strings (Vidal et al., 2010). In successful memory encoding, words later recalled show lower low-frequency power and higher HFA than words later forgotten (Sederberg et al., 2003; Long et al., 2014). This broadband tilt also appears outside memory, during arithmetic (Randazzo et al., 2019), where it predicts response speed, mainly in frontal regions (DeHaan et al., 2025). Not every region is engaged by a stimulus, however – alpha increases in task-irrelevant cortex are thought to actively inhibit it (Jensen and Mazaheri, 2010), and HG often decreases during tasks in default-mode regions (Ossandón et al., 2011).

Timing of visual word processing

Reading a word unfolds in stages: visual features are extracted within about 100 ms (Thorpe et al., 1996), the letter string is recognised as a word at about 200 ms (Cohen et al., 2000), its sound is activated automatically (van Orden, 1987), and its meaning is integrated with context at about 400 ms (Kutas and Hillyard, 1980). These stages map onto a reading network that runs from occipital cortex through left fusiform gyrus, which contains the visual word form area (VWFA; Cohen et al., 2000; Dehaene and Cohen, 2011), to temporal, supramarginal and inferior frontal cortex (Price, 2012; Taylor et al., 2013), with posterior responses to letter strings preceding more anterior responses to lexical status (Nobre et al., 1994; Allison et al., 1994).

iEEG resolves when each part of this network responds. Gamma activity begins at about 150 ms in basal temporal cortex, reaches the VWFA at about 200 ms, and reaches superior temporal and Broca's areas after 400 ms (Mainy et al., 2008); word-selective ventral temporal electrodes respond at about 150 ms (Kadipasaoglu et al., 2016); and the effects of word properties move anteriorly along the ventral stream, with feedback to posterior regions at 250–500 ms (Woolnough et al., 2021). Beyond the ventral stream, MEG shows responses proceeding from temporal cortex to Broca's area and finally to orbitofrontal cortex at about 370 ms (Halgren et al., 2002), and MTL neurons respond later still, at about 270–400 ms (Mormann et al., 2008). Early visual responses are sharp, whereas later responses are broad, consistent with feedforward input followed by feedback (Halgren et al., 2006; Ghuman et al., 2014). However, most intracranial studies of reading record from tens of patients with sparse, clinically determined coverage, and focus on the ventral stream, so the order in which regions across the whole brain respond to a word has not been measured within one large dataset.

Synchrony between regions

Beyond local power, word onset may change how regions synchronize. Alpha phase synchrony is thought to coordinate distant regions (Palva and Palva, 2007), and alpha power can fall while alpha synchrony between regions rises (Hanslmayr et al., 2016). In free recall, slow-frequency synchrony increases while high-frequency activity desynchronizes during successful encoding and retrieval (Burke et al., 2013; Solomon et al., 2017), and theta hubs mark successful retrieval in this dataset (Rao et al., 2025). Connectivity measured at stimulus onset is harder to interpret than power: an evoked response shared across regions can raise phase and envelope coupling without any interaction between them, and subtracting the average evoked response does not fully remove it, because its amplitude and latency vary across trials (Truccolo et al., 2002; David et al.,

2006). We therefore treat synchrony as a secondary analysis.

Temporal expectation

Reaction times and neural responses depend on how long the participant waits for a stimulus and on the changing probability that it will appear (the hazard rate; Niemi and Näätänen, 1981; Nobre and van Ede, 2018). Temporally expected events evoke larger visual responses: in monkey V1, gamma increases and alpha decreases when a stimulus is expected (Lima et al., 2011), and expectation speeds and enhances visual processing in humans (Doherty et al., 2005).

Predicting recall from the word-onset response

Spectral features recorded while a word is studied predict whether it will be recalled (L2-regularised logistic regression; Ezzyat et al., 2017, 2018), although these classifiers partly track general fluctuations in brain state rather than encoding of the item (Weidemann and Kahana, 2021). Activity before a stimulus can also predict memory: pre-stimulus hippocampal theta predicts later recognition but not recall (Merkow et al., 2014), and pre-stimulus MTL and cortical low-frequency activity predict encoding success in iEEG (Fell et al., 2011). Comparing recalled with forgotten words region by region, HFA predicting later recall appears first in visual, fusiform and medial temporal regions (about 0–500 ms) and later in left inferior frontal, posterior parietal and ventrolateral temporal cortex (about 400–900 ms; Burke et al., 2014). These approaches either pool features across the brain and the whole word period, or test each region and time point for a difference in mean activity; how well each region's single-trial response predicts recall, and when that prediction arises, has not been measured with classifiers.

The present study

There have been few whole-brain maps of the response to word onset itself, contrasting each word to a pre-stimulus baseline. With 362 patients, we obtain a complete set of HG onset latencies across all regions, extending the existing literature. We also test phase (alpha ciPLV) and envelope (HG AEC-c) coupling in the same data.

Methods

Data and cohort

- OpenNeuro BIDS releases of the RAM / Penn free-recall data (FR1, catFR1) and pyFR, bipolar montage. OpenNeuro ds004789 (FR1), ds004809 (catFR1), ds004865 (pyFR).
- 362 subjects, 1009 sessions: FR1 232 subj / 466 sess, catFR1 212 / 465, pyFR 38 / 78.
- Bipolar channels whose contacts are > 20 mm apart, and flat channels are dropped.
- Exclusions: missing / < 499 Hz sampling rate, denylist / manual, mains-harmonic peak $> 5\times$ its neighbours at a non-notched frequency (9 sessions, all 100 Hz peaks at 60 Hz sites).
- Contrast: **word on** (0–600 ms after word onset) vs **word off** (-700 to -100 ms, the blank screen before the same word). Words on screen 1600 s; blank between words ~750-1250 ms.

Preprocessing

- Word-off and word-on clips loaded separately per word, each with 500 ms of real data buffer on both sides.
- Resampled to 500 Hz, notch 60 Hz + 120 Hz harmonic (4th-order Butterworth, ± 2 Hz); buffer cropped before every estimate. (Widmann et al., 2015; de Cheveigné and Nelken, 2019; Gramfort, 2013; Virtanen et al., 2020)

Electrode localization

- Bipolar pairs labelled by lab localization default (depth: volumetric; grid/strip: surface). (Desikan et al. 2006; Fischl et al. 2002, 2004; Avants et al. 2008; Wang et al. 2013; Yushkevich et al. 2015; Azarion et al. 2014 (check); bipolar: Mercier et al. 2022; Li et al. 2018)
- Depth pairs with no volumetric label in the BIDS release: label recovered from the RAM imaging pipeline's label files (pair file, or the midpoint contact for skip pairs; white matter / ventricles stay unlabelled); the 13% without a recoverable label get the majority label of expert-labelled depth pairs of other subjects within 5 mm (MNI).
- 12 Burke ROIs: frontal, temporal, parietal, occipital, limbic, hippocampus × hemisphere; sub regions = anatomical labels within them. (Burke et al., 2013)

Power

- Alpha (8–13 Hz) and 5–150 Hz spectrum: multitaper, 2 Hz bandwidth, 600 ms windows; spectrum in 11 log-spaced bins (50, 60, 100, 120 Hz $\pm$ 2 Hz excluded). (Thomson, 1982; Slepian, 1978; Percival and Walden, 1993; Prerau et al., 2017)
- High gamma (70–150 Hz): Hilbert in 8×10 Hz sub-bands, each divided by its word-off mean, averaged (broadband, every frequency weighted equally). (Edwards et al., 2009; Bruns, 2004; Manning et al., 2009; Ray and Maunsell, 2011) **(check Edwards et al. 2009)**
- Per electrode: Cohen's d of per-trial power, word on vs word off. (Cohen, 1988; Lakens, 2013)

High-gamma latency

- Responsive electrodes: per-trial high-gamma power word on $>$ word off (one-tailed Welch t test, BH-FDR within session): 4950 of 122,339

electrode-sessions, 299 subjects. (WELCH, 1947; Benjamini and Hochberg, 1995); on selecting electrodes and measuring them in the same data, see Kriegeskorte et al. (2009)
- Latency: trial-mean envelope minus word-off mean, 30 ms smoothing, first time it reaches half its peak within 0–800 ms (half-maximum). (Kiesel et al., 2008; Liesefeld, 2018); on filter smearing of onsets, see VanRullen (2011)
- Subject median per region → mean ± SEM across subjects; 12 ROIs and subregions (hemispheres pooled).

Synchrony

- Alpha: ciPLV (MNE `spectral_connectivity_epochs`, multitaper, 2 Hz bandwidth). (Bruña et al. 2018; Nolte et al. 2004; MNE-Connectivity (Zenodo))
- High gamma: AEC-c (70–110 Hz band-pass + Hilbert, `envelope_correlation`, pairwise orthogonalized, signed, mean over trials); 70–110 Hz so the envelope stays below the band (Bedrosian). (Hipp et al., 2012; Bedrosian, 1963)
- Pairs sharing a contact excluded. (Mercier et al., 2017) Per electrode: mean on − off change over all partners (Rao et al., 2025).

Time-resolved network (epochs)

- 9 overlapping 200 ms epochs every 100 ms, 0–1000 ms (stops before the word's offset response at 1600 ms), each minus the mean of three 200 ms word-off epochs.
- Region pairs with ≥ 100 subjects. Hub (Rao): a region's mean change over its connections; t across subjects; two-stage FDR (Benjamini–Krieger–Yekutieli) over region × epoch. (Benjamini et al., 2006; Rubinov and Sporns, 2010)
- Edges: each significant hub's 5 strongest connections (|mean change|), either sign; width ∝ |t| (Solomon et al., 2017 style).

Mixed-effects models (R lme4 / emmeans; `mixed_models.R`)

- Latency: latency_ms ~ lobe + (1|sub) + (1|sub:lobe) + (1|sub:region), one row per responsive electrode; all 15 lobe pairs, Holm.
- Power: cohens_d ~ lobe × band + (1 + band|sub) + (1|sub:lobe) + (1|sub:lobe:band) + (1|sub:region) + (1|sub:label); lobe × band interaction contrast per lobe pair, Holm.

Anticipation

- Responsive electrodes (as latency); trials split at the session's median blank length; peak (0–800 ms) of the trial-mean high-gamma response after long minus short waits; subject mean per ROI, one-sample t, BH-FDR over ROIs.

Recall prediction (encoding contrast: recalled vs not recalled)

- High-gamma response (as latency) in 100 ms windows, 0–1000 ms, z-scored within session; per subject × lobe (≥ 5 electrodes) × window: L2 logistic regression (C = 7.2e-4, Ezzyat et al., 2017), balanced class weights, leave-one-session-out (single-session subjects: leave-one-list-out); AUC of pooled held-out predictions.
- Across subjects: one-sided t of AUC > 0.5, BH-FDR over lobe × window; permutation chance (~0.49) reported only.

Reliability

- Subjects' sessions split at random into two halves (≥ 2 sessions per half; riley-thesis rules); per half, d (power) or node strength (synchrony) averaged per anatomical region; test-retest = Pearson r across the subject's regions between halves.

Statistics

- Sessions averaged per electrode, electrodes per subject × region (≥ 1 electrode); one-sample t vs 0 across subjects; BH-FDR within each figure / band; ≥ 30 subjects per region. (Holmes and Friston, 1998; Benjamini and Hochberg, 1995; Seabold and Perktold, 2010)

Results

1. Power changes at word onset

- Alpha: decrease in occipital (L −0.13, R −0.10 d), smaller decreases in hippocampus (L −0.016, R −0.025), R limbic (−0.023) and R temporal (−0.017); increase in frontal (L +0.031, R +0.023) and L parietal (+0.024).
- High gamma: large increase in occipital (L +0.40, R +0.29 d); smaller increases in limbic (+0.03–0.04), hippocampus (+0.03–0.05), L temporal and L parietal (+0.02); no change in frontal.
- Occipital subregions (hemispheres pooled; the 4 with most subjects): fusiform (221), lateral occipital (100), lingual (89), cuneus (37). Fusiform, lateral occipital, lingual: alpha/beta decrease and gamma increase (≥37–51 Hz), the same pattern as occipital cortex as a whole; fusiform strongest (t ≈ −7.8 at 11 Hz, +8.7 at 69 Hz); lateral occipital also a 6 Hz increase; cuneus weakest (gamma at 51–69 Hz only). TFCE, family-wise $p < .05$. (SMITH and NICHOLS, 2009; Mensen and Khatami, 2013; Nichols and Holmes, 2002)

Spectral profile

- Frontal: theta/low-alpha increase (t up to 11.7 at 6 Hz) with beta decrease (t to −13.4 at 27 Hz), and a small high-gamma increase at 129 Hz (t 3.8).
- Occipital: alpha–beta decrease (11 Hz t ≈ −7.6), broadband increase ≥ 37 Hz (t up to 8.7).

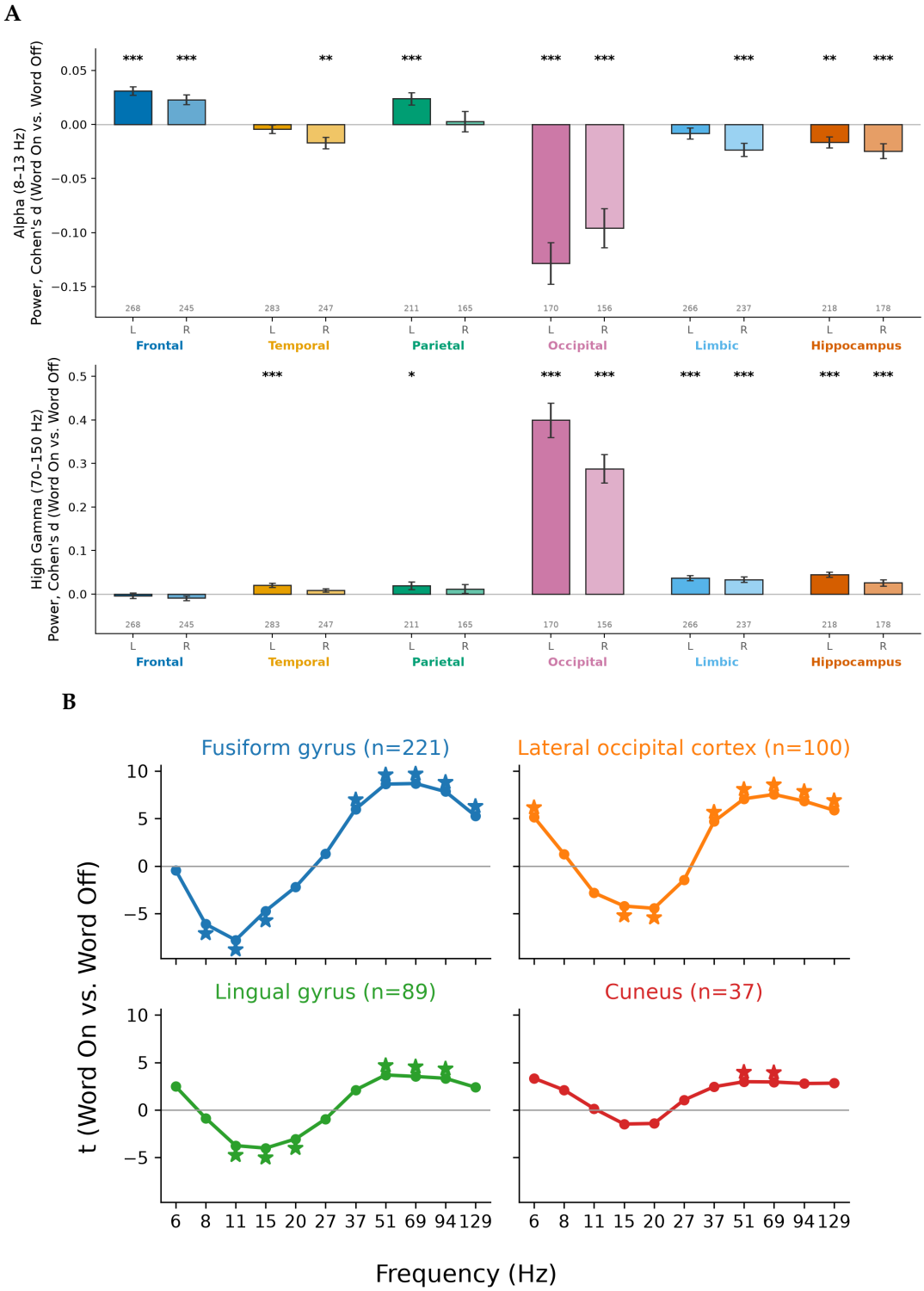


Figure 1. Power changes at word onset (word on vs. word off). (A) Cohen's d of alpha (8–13 Hz, top) and high-gamma (70–150 Hz, bottom) power per region; mean ± SEM across subjects; numbers: subjects; stars: BH-FDR. (B) The strongest region, occipital cortex, by subregion: t across subjects per frequency for the four occipital subregions with the most subjects (hemispheres pooled); stars: significant (TFCE, family-wise $p < .05$).

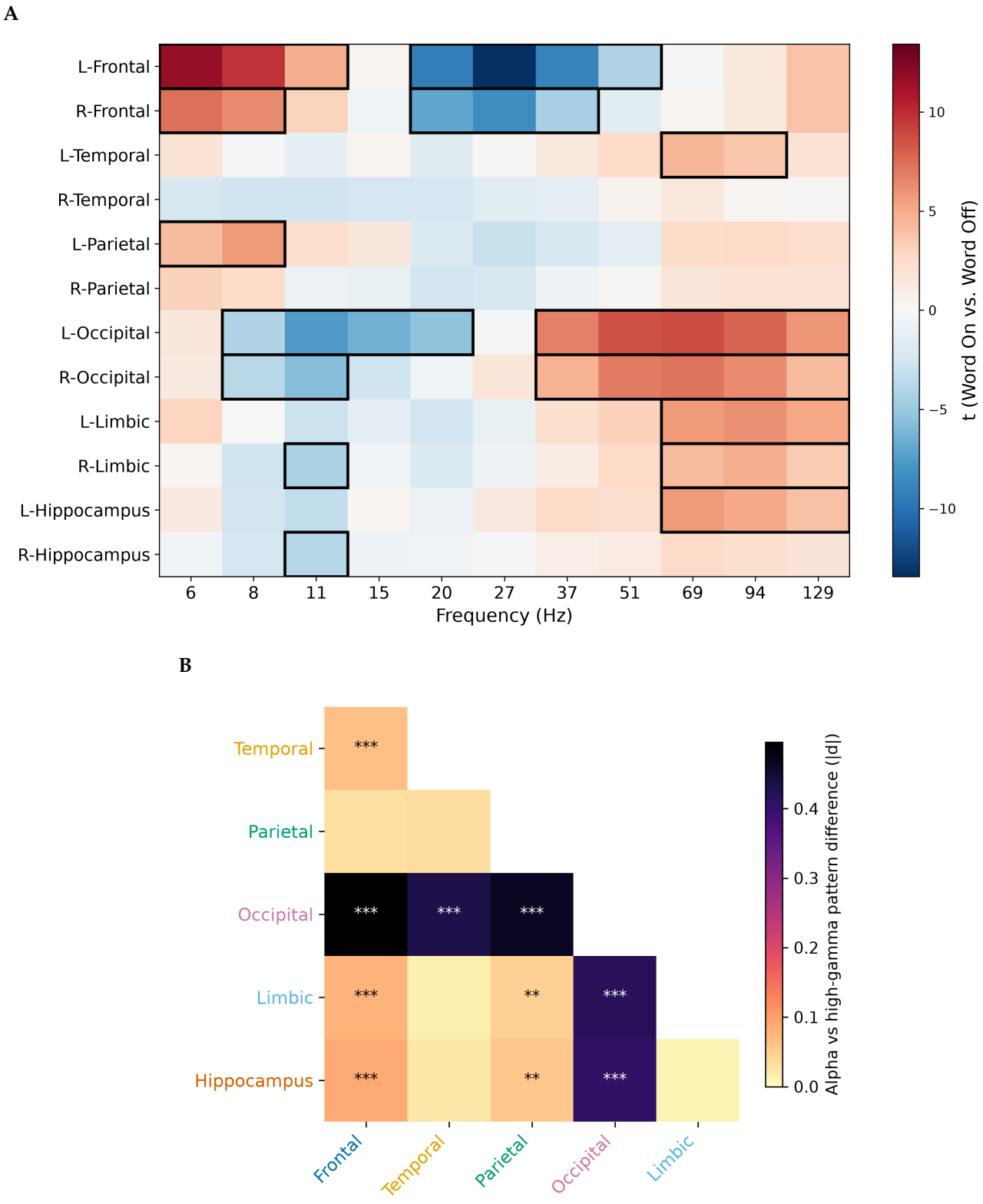


Figure 2. Spectral profile of the word-onset response. (A) t across subjects per region and frequency (5–150 Hz, log-spaced bins; 50, 60, 100, 120 Hz excluded); outlines: significant runs (TFCE, family-wise $p < .05$). (B) Mixed-effects lobe × band interaction: how much the alpha-vs-high-gamma pattern differs between each pair of lobes ($|d|$); stars: Holm-corrected p.

- Limbic / hippocampus: mild alpha decrease, gamma increase.

Mixed model (lobe × band)

- Does the alpha-vs-high-gamma pattern differ between lobes (lobe × band interaction)? 10 of 15 pairs differ (Holm).
- Occipital differs from every other lobe (|d| 0.41–0.50); frontal from temporal, limbic and hippocampus (0.07–0.09); parietal from limbic and hippocampus.
- Not different: frontal vs parietal ($p = .053$), parietal vs temporal ($p = .050$), limbic vs temporal, hippocampus vs limbic, hippocampus vs temporal.

2. Order of activation

- Posterior → anterior: R occipital 132 ms, L occipital 156 → limbic (~164–166), R temporal 171, R parietal 177 → L hippocampus 201, L temporal 203, R hippocampus 210 → R frontal 222, L parietal 227 → L frontal 245 ms.
- Subregions (≥ 30 subjects): lateral occipital 123 → fusiform 156, lingual 158 → parahippocampal 168, superior parietal 175 → inferior / middle temporal 183 / 195, hippocampus 203, superior temporal 209 → middle frontal 213, supramarginal 228, precentral 229, inferior frontal 238 → orbitofrontal 316 ms.
- Early regions: sharp peak ~150 ms; hippocampus / insula / frontal: broad peaks ~350–550 ms.

Mixed model (latency per lobe)

- Estimated latency: occipital 150 → limbic 170 → temporal 193 → hippocampus 209 ≈ parietal 210 → frontal 238 ms.
- Neighbouring steps (brackets): only parietal < frontal ($p = .024$); occipital–limbic ($p = .14$), limbic–temporal ($p = .10$), temporal–hippocampus ($p = .38$), hippocampus–parietal ($p = .93$) not separable.
- All 15 pairs (CSV): 9 differ (Holm $p < .05$), e.g. occipital before temporal, parietal, hippocampus and frontal; frontal after all but hippocampus ($p = .10$, the least precise lobe: 67 subjects). Posterior → anterior holds across two or more steps, not every neighbouring step.

3. Synchrony

- Alpha ciPLV increases in 9 / 12 ROIs (not L / R hippocampus, R limbic), about twice as much in occipital (~0.0015–0.0016) as elsewhere (~0.0003–0.0009).
- High-gamma AEC-c: no significant change in any ROI.

Time-resolved network

- Alpha: 69 hub × window cells, all positive. Occipital hubs in the first four windows (0–500 ms, both hemispheres) and once at 500–700 ms; frontal, limbic and temporal hubs from the first window onward; parietal hubs from the 100–300 ms window onward, in both hemispheres from 200 ms; hippocampus at 100–400 ms. Fine regions (8 cells): L fusiform 0–400 ms, then L middle temporal 400–1000 ms.
- High gamma (not shown): 2 hubs (L occipital 0–200 ms, L limbic 200–400 ms).

4. Anticipation (blank length before the word)

- Occipital high-gamma peak larger after long than short waits (L +0.072, $q = .022$; R +0.049, $q = .029$); no other ROI significant.
- Consistent with temporal expectation boosting visual responses (Lima et al., 2011; Nobre and van Ede, 2018, monkey V1), but a shorter blank also leaves less time for the previous word's offset response to decay (baseline).

5. Recall prediction over the response (encoding contrast)

- No lobe predicts recall before 200 ms (first significant window: occipital 200–300 ms; frontal from 300 ms).
- Frontal: sustained 300–1000 ms, highest at 800–900 ms (AUC 0.525).
- Occipital: 200–700 ms only (peak 0.519 at 500–600 ms); temporal 400–900 ms; parietal 300–500 and 600–800 ms; hippocampus 500–600 ms; limbic 600–700 ms.
- AUCs are low (≤ 0.525) because each classifier uses one lobe's high gamma in one window. Same direction as Burke, Long et al. 2014 (encoding contrast: early posterior, late frontal HFA SME).

6. Reliability across sessions

- 85 subjects with ≥ 2 sessions per half. Power maps repeat: median r 0.67 (alpha), 0.81 (high gamma). Synchrony maps much less: 0.19 (alpha ciPLV), 0.14 (high-gamma AEC-c).

Limitations / discussion

- Word-off baseline starts ~80-410 ms after the previous word's offset: possible offset response / anticipation in the baseline; the blank-length split (Results 4) cannot separate the two.
- No ERP subtraction: evoked activity contributes to power and synchrony, especially occipital.
- Neighbour-transferred labels (~11% / 21% disagreement in LOSO) only for the 13% of unlabelled depth pairs the imaging-pipeline label files lack.

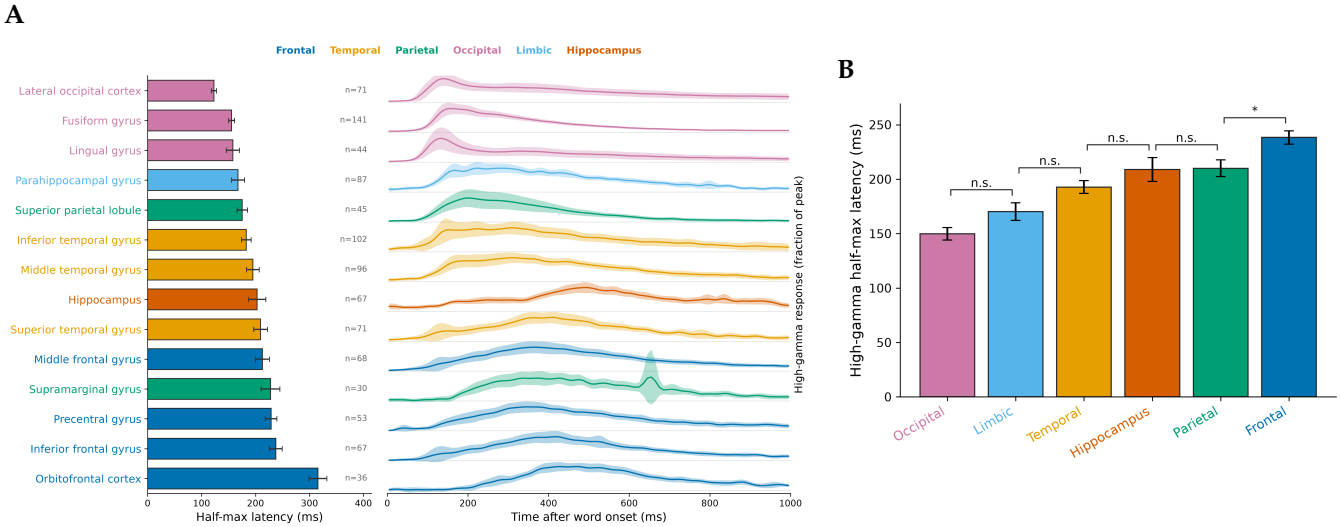


Figure 3. Order of high-gamma responses after word onset. (A) Half-max latency per anatomical subregion (left; mean ± SEM across subjects; n: subjects) and the average high-gamma response over time, as a fraction of its peak (right). (B) Estimated latency per lobe from a mixed-effects model (mean ± SE); brackets: neighbouring lobes in the latency order (Holm over all 15 pairs).

- High-gamma phase metrics avoided (broadband HFA has no meaningful phase); AEC-c null is consistent with high gamma being local and asynchronous (Solomon et al., 2017, 2019).
- Multiple comparisons corrected within figures, not across.

Stimulus-locked activity in connectivity

- Subtracting the average ERP leaves a residual because amplitude and latency vary across trials: Truccolo et al., 2002; Wang et al., 2008; Knuth et al., 2006.
- Evoked and induced activity can't be cleanly separated: David et al., 2006; Shah, 2004. This supports the PI's position.
- Is alpha phase reset? For: Makeig et al., 2002; Hanslmayr et al., 2007. Against: Mazaheri and Jensen, 2006. Hard to tell apart: Yeung et al., 2004; Sauseng et al., 2007.
- Task-evoked activity inflates connectivity: Cole et al., 2019; van Diepen and Mazaheri, 2018; Kramer et al., 2008 (sharp transients).
- Alternatives to subtraction:
  - Matched condition contrast with a label-shuffle null: Solomon et al., 2017; Maris et al., 2007.
  - Trial-shuffled (shift-predictor) surrogates: Perkel et al., 1967; Aertsen et al., 1989; Brody, 1999.
  - Connectivity aligned to internal HFA peaks: Dede et al., 2025.
  - Phase consistency on the same data: Colyer and Kahana, 2025.

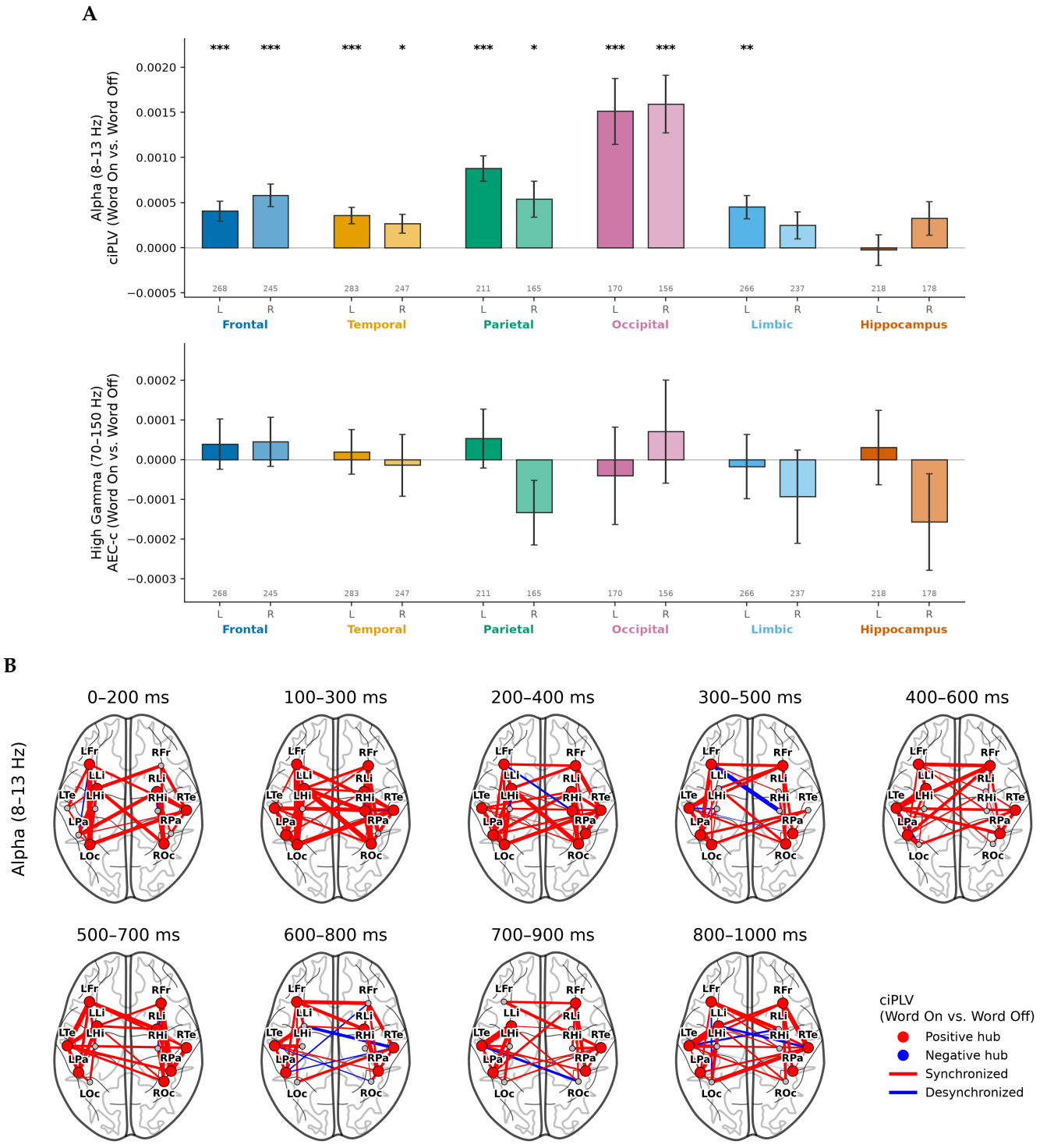


Figure 4. Synchrony at word onset. (A) Change in alpha ciPLV (top) and high-gamma AEC-c (bottom) per region, word on vs. off: each electrode's mean change over all partners, averaged per subject; mean ± SEM across subjects; stars: BH-FDR. (B) Time-resolved alpha ciPLV network in 200 ms windows. Red nodes: hubs, regions whose mean change over their connections differs from zero (two-stage FDR over regions × windows); lines: each hub's five strongest connections (red: synchronized, blue: desynchronized; width ∝ $|t|$).

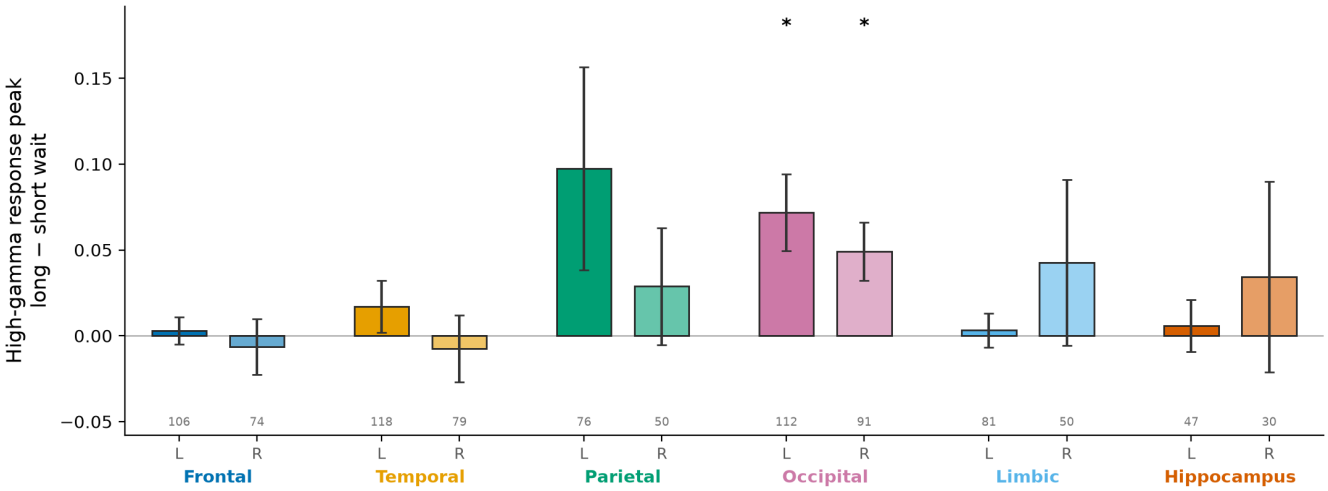


Figure 5. High-gamma response peak after long minus short waits (median split of the blank before each word), responsive electrodes only; mean ± SEM across subjects; stars: BH-FDR.

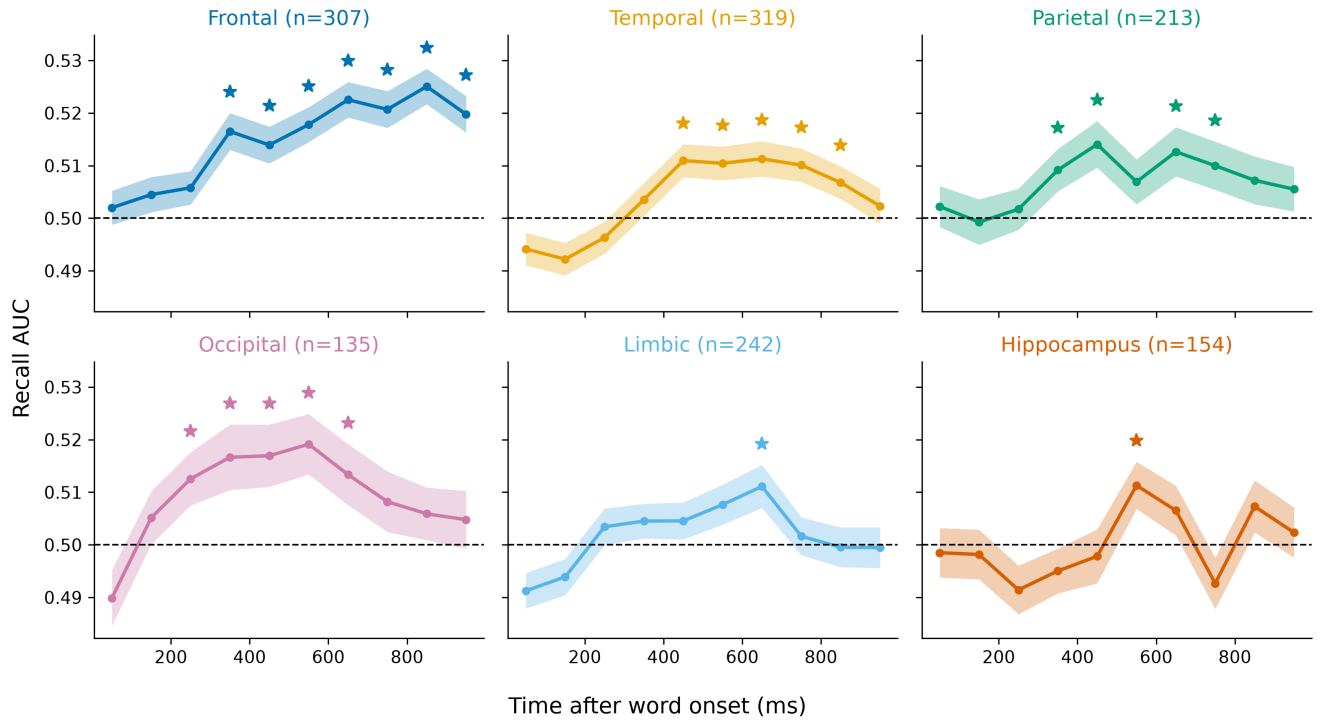


Figure 6. Recall prediction from the high-gamma response (encoding contrast: recalled vs. not recalled). AUC per lobe and 100 ms window, mean ± SEM across subjects; dashed line: 0.5; stars: AUC > 0.5, BH-FDR over lobe × window.

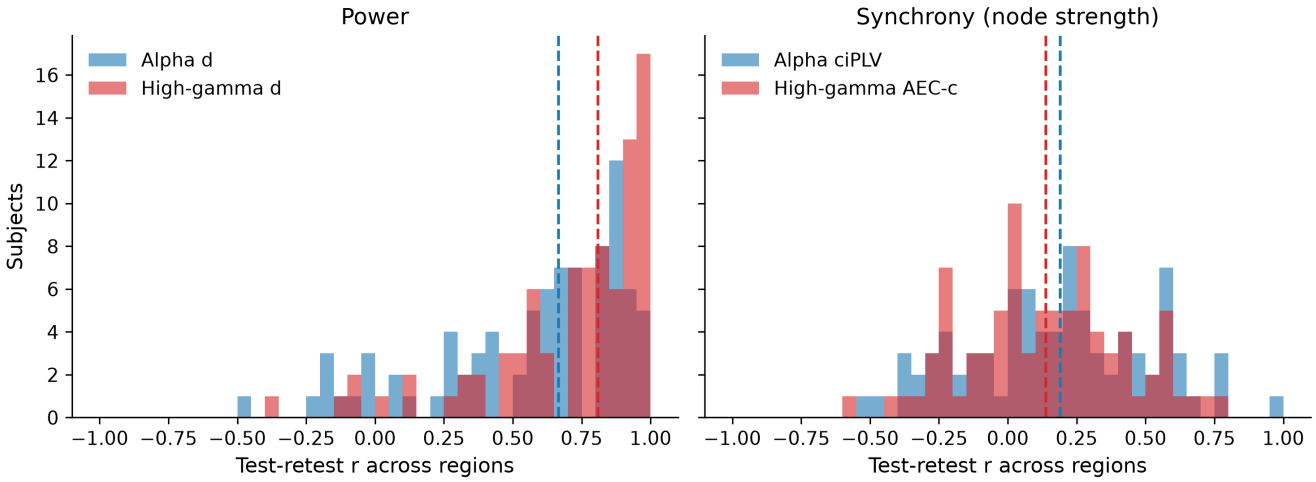


Figure 7. Session-to-session reliability: per subject, Pearson r across regions between two random halves of the subject's sessions, for power (left) and synchrony node strength (right); dashed lines: medians.

References

Aertsen AM, Gerstein GL, Habib MK, Palm G (1989) Dynamics of neuronal firing correlation: modulation of "effective connectivity". *Journal of Neurophysiology* 61:900–917.

Allison T, McCarthy G, Nobre A, Puce A, Belger A (1994) Human extrastriate visual cortex and the perception of faces, words, numbers, and colors. *Cerebral Cortex* 4:544–554.

Bedrosian E (1963) A product theorem for hilbert transforms. *Proceedings of the IEEE* 51:868–869.

Benjamini Y, Hochberg Y (1995) Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society Series B: Statistical Methodology* 57:289–300.

Benjamini Y, Krieger AM, Yekutieli D (2006) Adaptive linear step-up procedures that control the false discovery rate. *Biometrika* 93:491–507.

Brody CD (1999) Correlations without synchrony. *Neural Computation* 11:1537–1551.

Bruns A (2004) Fourier-, hilbert- and wavelet-based signal analysis: are they really different approaches? *Journal of Neuroscience Methods* 137:321–332.

Burke JF, Long NM, Zaghloul KA, Sharan AD, Sperling MR, Kahana MJ (2014) Human intracranial high-frequency activity maps episodic memory formation in space and time. *NeuroImage* 85:834–843.

Burke JF, Ramayya AG, Kahana MJ (2015) Human intracranial high-frequency activity during memory processing: neural oscillations or stochastic volatility? *Current Opinion in Neurobiology* 31:104–110.

Burke JF, Zaghloul KA, Jacobs J, et al. (2013) Synchronous and asynchronous theta and gamma activity during episodic memory formation. *The Journal of Neuroscience* 33:292–304.

Cohen J (1988) *Statistical Power Analysis for the Behavioral Sciences* Routledge.

Cohen L, Dehaene S, Naccache L, et al. (2000) The visual word form area. *Brain* 123:291–307.

Cole MW, Ito T, Schultz D, Mill R, Chen R, Cocuzza C (2019) Task activations produce spurious but systematic inflation of task functional connectivity estimates. *NeuroImage* 189:1–18.

Colyer RA, Kahana MJ (2025) Phase consistency dynamics of memory encoding. *The Journal of Neuroscience* 45:e2077242025.

Crone N (1998a) Functional mapping of human sensorimotor cortex with electrocorticographic spectral analysis. i. alpha and beta event- related desynchronization. *Brain* 121:2271–2299.

Crone N (1998b) Functional mapping of human sensorimotor cortex with electrocorticographic spectral analysis. ii. event-related synchronization in the gamma band. *Brain* 121:2301–2315.

David O, Kilner JM, Friston KJ (2006) Mechanisms of evoked and induced responses in meg/eeg. *NeuroImage* 31:1580–1591.

de Cheveigné A, Nelken I (2019) Filters: When, why, and how (not) to use them. *Neuron* 102:280–293.

Dede AJO, Cross ZR, Gray SM, et al. (2025) Episodic memory involves transient and sparse connectivity aligned to both internal and external events. *PLOS Biology* 23:e3003481.

DeHaan RD, Ezzyat Y, Randazzo MJ, et al. (2025) Neural components underlying successful free recall are specific to episodic memory. *bioRxiv* (*preprint*) .

Dehaene S, Cohen L (2011) The unique role of the visual word form area in reading. *Trends in Cognitive Sciences* 15:254–262.

Doherty JR, Rao A, Mesulam MM, Nobre AC (2005) Synergistic effect of combined temporal and spatial expectations on visual attention. *The Journal of Neuroscience* 25:8259–8266.

Edwards E, Soltani M, Kim W, et al. (2009) Comparison of time–frequency responses and the event-related potential to auditory speech stimuli in human cortex. *Journal of Neurophysiology* 102:377–386.

Ezzyat Y, Kragel JE, Burke JF, et al. (2017) Direct brain stimulation modulates encoding states and memory performance in humans. *Current Biology* 27:1251–1258.

Ezzyat Y, Wanda PA, Levy DF, et al. (2018) Closed-loop stimulation of temporal cortex rescues functional networks and improves memory. *Nature Communications* 9.

Fell J, Ludowig E, Staresina BP, et al. (2011) Medial temporal theta/alpha power enhancement precedes successful memory encoding: Evidence based on intracranial eeg. *The Journal of Neuroscience* 31:5392–5397.

Ghuman AS, Brunet NM, Li Y, et al. (2014) Dynamic encoding of face information in the human fusiform gyrus. *Nature Communications* 5.

Gramfort A (2013) Meg and eeg data analysis with mne-python. *Frontiers in Neuroscience* 7.

Halgren E, Dhond RP, Christensen N, et al. (2002) N400-like magnetoencephalography responses modulated by semantic context, word frequency, and lexical class in sentences. *NeuroImage* 17:1101–1116.

Halgren E, Wang C, Schomer DL, et al. (2006) Processing stages underlying word recognition in the anteroventral temporal lobe. *NeuroImage* 30:1401–1413.

Hanslmayr S, Aslan A, Staudigl T, Klimesch W, Herrmann CS, Bäuml KH (2007) Prestimulus oscillations predict visual perception performance between and within subjects. *NeuroImage* 37:1465–1473.

Hanslmayr S, Staresina BP, Bowman H (2016) Oscillations and episodic memory: Addressing the synchronization/desynchronization conundrum. *Trends in Neurosciences* 39:16–25.

Hipp JF, Hawellek DJ, Corbetta M, Siegel M, Engel AK (2012) Large-scale cortical correlation structure of spontaneous oscillatory activity. *Nature Neuroscience* 15:884–890.

Holmes A, Friston K (1998) Generalisability, random effects & population inference. *NeuroImage* 7:S754.

Jensen O, Mazaheri A (2010) Shaping functional architecture by oscillatory alpha activity: Gating by inhibition. *Frontiers in Human Neuroscience* 4.

Kadipasaoglu CM, Conner CR, Whaley ML, Baboyan VG, Tandon N (2016) Category-selectivity in human visual cortex follows cortical topology: A grouped iceeg study. *PLOS ONE* 11:e0157109.

Kiesel A, Miller J, Jolicœur P, Brisson B (2008) Measurement of erp latency differences: A comparison of single-participant and jackknife-based scoring methods. *Psychophysiology* 45:250–274.

Knuth KH, Shah AS, Truccolo WA, Ding M, Bressler SL, Schroeder CE (2006) Differentially variable component analysis: Identifying multiple evoked components using trial-to-trial variability. *Journal of Neurophysiology* 95:3257–3276.

Kramer MA, Tort AB, Kopell NJ (2008) Sharp edge artifacts and spurious coupling in eeg frequency comodulation measures. *Journal of Neuroscience Methods* 170:352–357.

Kriegeskorte N, Simmons WK, Bellgowan PSF, Baker CI (2009) Circular analysis in systems neuroscience: the dangers of double dipping. *Nature Neuroscience* 12:535–540.

Kucewicz MT, Saboo K, Berry BM, et al. (2019) Human verbal memory encoding is hierarchically distributed in a continuous processing stream. *eneuro* 6:ENEURO.0214–18.2018.

Kutas M, Hillyard SA (1980) Reading senseless sentences: Brain potentials reflect semantic incongruity. *Science* 207:203–205.

Lachaux JP, George N, Tallon-Baudry C, et al. (2005) The many faces of the gamma band response to complex visual stimuli. *NeuroImage* 25:491–501.

Lakens D (2013) Calculating and reporting effect sizes to facilitate cumulative science: a practical primer for t-tests and anovas. *Frontiers in Psychology* 4.

Liesefeld HR (2018) Estimating the timing of cognitive operations with meg/eeg latency measures: A primer, a brief tutorial, and an implementation of various methods. *Frontiers in Neuroscience* 12.

Lima B, Singer W, Neuenschwander S (2011) Gamma responses correlate with temporal expectation in monkey primary visual cortex. *The Journal of Neuroscience* 31:15919–15931.

Long NM, Burke JF, Kahana MJ (2014) Subsequent memory effect in intracranial and scalp eeg. *NeuroImage* 84:488–494.

Mainy N, Jung J, Baciu M, et al. (2008) Cortical dynamics of word recognition. *Human Brain Mapping* 29:1215–1230.

Makeig S, Westerfield M, Jung TP, et al. (2002) Dynamic brain sources of visual evoked responses. *Science* 295:690–694.

Manning JR, Jacobs J, Fried I, Kahana MJ (2009) Broadband shifts in local field potential power spectra are correlated with single-neuron spiking in humans. *The Journal of Neuroscience* 29:13613–13620.

Maris E, Schoffelen JM, Fries P (2007) Nonparametric statistical testing of coherence differences. *Journal of Neuroscience Methods* 163:161–175.

Marks VS, Saboo KV, Topçu Ç, et al. (2021) Independent dynamics of low, intermediate, and high frequency spectral intracranial eeg activities during human memory formation. *NeuroImage* 245:118637.

Mazaheri A, Jensen O (2006) Posterior α activity is not phase-reset by visual stimuli. *Proceedings of the National Academy of Sciences* 103:2948–2952.

Mensen A, Khatami R (2013) Advanced eeg analysis using threshold-free cluster-enhancement and non-parametric statistics. *NeuroImage* 67:111–118.

Mercier MR, Bickel S, Megevand P, et al. (2017) Evaluation of cortical local field potential diffusion in stereotactic electroencephalography recordings: A glimpse on white matter signal. *NeuroImage* 147:219–232.

Merkow MB, Burke JF, Stein JM, Kahana MJ (2014) Prestimulus theta in the human hippocampus predicts subsequent recognition but not recall. *Hippocampus* 24:1562–1569.

Mormann F, Kornblith S, Quiroga RQ, et al. (2008) Latency and selectivity of single neurons indicate hierarchical processing in the human medial temporal lobe. *The Journal of Neuroscience* 28:8865–8872.

Nichols TE, Holmes AP (2002) Nonparametric permutation tests for functional neuroimaging: A primer with examples. *Human Brain Mapping* 15:1–25.

Niemi P, Näätänen R (1981) Foreperiod and simple reaction time. *Psychological Bulletin* 89:133–162.

Nobre AC, Allison T, McCarthy G (1994) Word recognition in the human inferior temporal lobe. *Nature* 372:260–263.

Nobre AC, van Ede F (2018) Anticipated moments: temporal structure in attention. *Nature Reviews Neuroscience* 19:34–48.

Ossandón T, Jerbi K, Vidal JR, et al. (2011) Transient suppression of broadband gamma power in the default-mode network is correlated with task complexity and subject performance. *The Journal of Neuroscience* 31:14521–14530.

Palva S, Palva JM (2007) New vistas for α-frequency band oscillations. *Trends in Neurosciences* 30:150–158.

Percival DB, Walden AT (1993) *Spectral Analysis for Physical Applications* Cambridge University Press.

Perkel DH, Gerstein GL, Moore GP (1967) Neuronal spike trains and stochastic point processes. *Biophysical Journal* 7:419–440.

Pfurtscheller G, Lopes da Silva F (1999) Event-related eeg/meg synchronization and desynchronization: basic principles. *Clinical Neurophysiology* 110:1842–1857.

Prerau MJ, Brown RE, Bianchi MT, Ellenbogen JM, Purdon PL (2017) Sleep neurophysiological dynamics through the lens of multitaper spectral analysis. *Physiology* 32:60–92.

Price CJ (2012) A review and synthesis of the first 20years of pet and fmri studies of heard speech, spoken language and reading. *NeuroImage* 62:816–847.

Randazzo MJ, Ezzyat Y, Kahana MJ (2019) Spectral tilt underlies mathematical problem solving. *bioRxiv (preprint)* .

Rao AM, DeHaan RD, Kahana MJ (2025) Synchronous theta networks characterize successful memory retrieval. *The Journal of Neuroscience* 45:e1332242025.

Ray S, Maunsell JHR (2011) Different origins of gamma rhythm and high-gamma activity in macaque visual cortex. *PLoS Biology* 9:e1000610.

Rubinov M, Sporns O (2010) Complex network measures of brain connectivity: Uses and interpretations. *NeuroImage* 52:1059–1069.

Sauseng P, Klimesch W, Gruber W, Hanslmayr S, Freunberger R, Doppelmayr M (2007) Are event-related potential components generated by phase resetting of brain oscillations? a critical discussion. *Neuroscience* 146:1435–1444.

Seabold S, Perktold J (2010) Statsmodels: Econometric and statistical modeling with python In *Proceedings of the 9th Python in Science Conference*, SciPy, p. 92–96. SciPy.

Sederberg PB, Kahana MJ, Howard MW, Donner EJ, Madsen JR (2003) Theta and gamma oscillations during encoding predict subsequent recall. *The Journal of Neuroscience* 23:10809–10814.

Shah AS (2004) Neural dynamics and the fundamental mechanisms of event-related brain potentials. *Cerebral Cortex* 14:476–483.

Slepian D (1978) Prolate spheroidal wave functions, fourier analysis, and uncertainty-v: The discrete case. *Bell System Technical Journal* 57:1371–1430.

SMITH S, NICHOLS T (2009) Threshold-free cluster enhancement: Addressing problems of smoothing, threshold dependence and localisation in cluster inference. *NeuroImage* 44:83–98.

Solomon EA, Kragel JE, Sperling MR, et al. (2017) Widespread theta synchrony and high-frequency desynchronization underlies enhanced cognition. *Nature Communications* 8.

Solomon EA, Stein JM, Das S, et al. (2019) Dynamic theta networks in the human medial temporal lobe support episodic memory. *Current Biology* 29:1100–1111.e4.

Taylor JSH, Rastle K, Davis MH (2013) Can cognitive models explain brain activation during word and pseudoword reading? a meta-analysis of 36 neuroimaging studies. *Psychological Bulletin* 139:766–791.

Thomson D (1982) Spectrum estimation and harmonic analysis. *Proceedings of the IEEE* 70:1055–1096.

Thorpe S, Fize D, Marlot C (1996) Speed of processing in the human visual system. *Nature* 381:520–522.

Truccolo WA, Ding M, Knuth KH, Nakamura R, Bressler SL (2002) Trial-to-trial variability of cortical evoked responses: implications for the analysis of functional connectivity. *Clinical Neurophysiology* 113:206–226.

van Diepen RM, Mazaheri A (2018) The caveats of observing inter-trial phase-coherence in cognitive neuroscience. *Scientific Reports* 8.

van Orden GC (1987) A rows is a rose: Spelling, sound, and reading. *Memory & Cognition* 15:181–198.

VanRullen R (2011) Four common conceptual fallacies in mapping the time course of recognition. *Frontiers in Psychology* 2.

Vidal JR, Ossandón T, Jerbi K, et al. (2010) Category-specific visual responses: An intracranial study comparing gamma, beta, alpha, and erp response selectivity. *Frontiers in Human Neuroscience* 4.

Virtanen P, Gommers R, Oliphant TE, et al. (2020) Scipy 1.0: fundamental algorithms for scientific computing in python. *Nature Methods* 17:261–272.

Wang X, Chen Y, Ding M (2008) Estimating granger causality after stimulus onset: A cautionary note. *NeuroImage* 41:767–776.

Weidemann CT, Kahana MJ (2021) Neural measures of subsequent memory reflect endogenous variability in cognitive function. *Journal of Experimental Psychology: Learning, Memory, and Cognition* 47:641–651.

WELCH BL (1947) The generalization of 'student's' problem when several different population varlances are involved. *Biometrika* 34:28–35.

Widmann A, Schröger E, Maess B (2015) Digital filter design for electrophysiological data – a practical approach. *Journal of Neuroscience Methods* 250:34–46.

Woolnough O, Donos C, Rollo PS, et al. (2021) Spatiotemporal dynamics of orthographic and lexical processing in the ventral visual pathway. *Nature Human Behaviour* 5:389–398.

Yeung N, Bogacz R, Holroyd CB, Cohen JD (2004) Detection of synchronized oscillations in the electroencephalogram: An evaluation of methods. *Psychophysiology* 41:822–832.

Supplementary

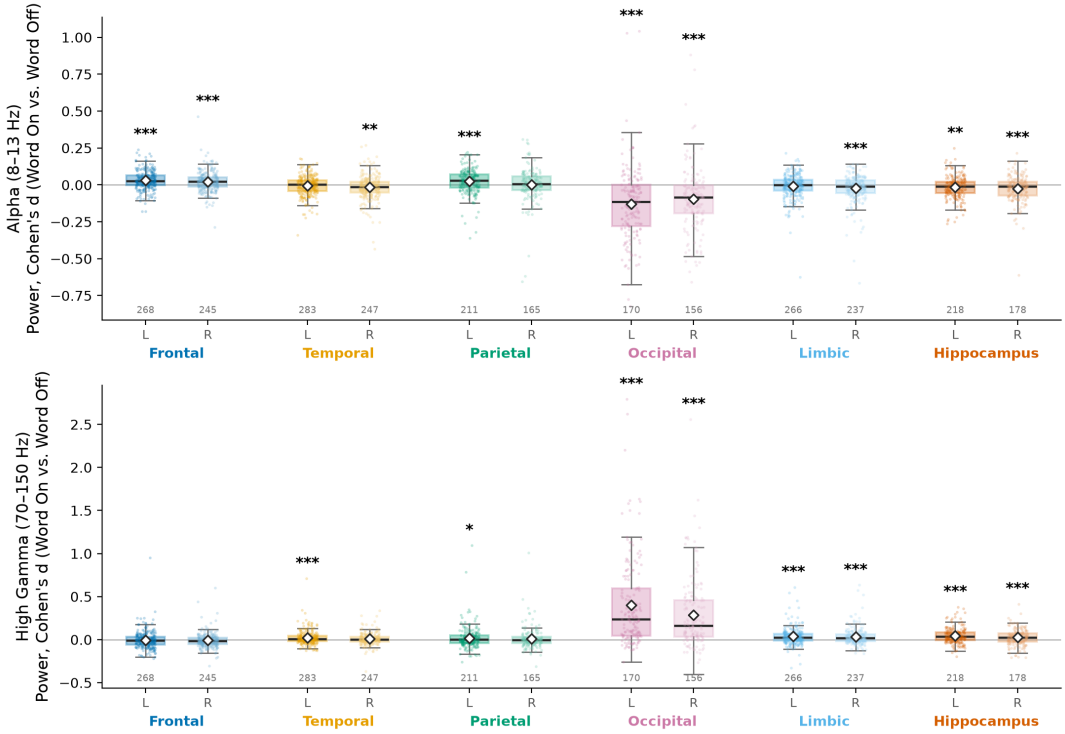


Figure 8. Power change per region, subject distributions.

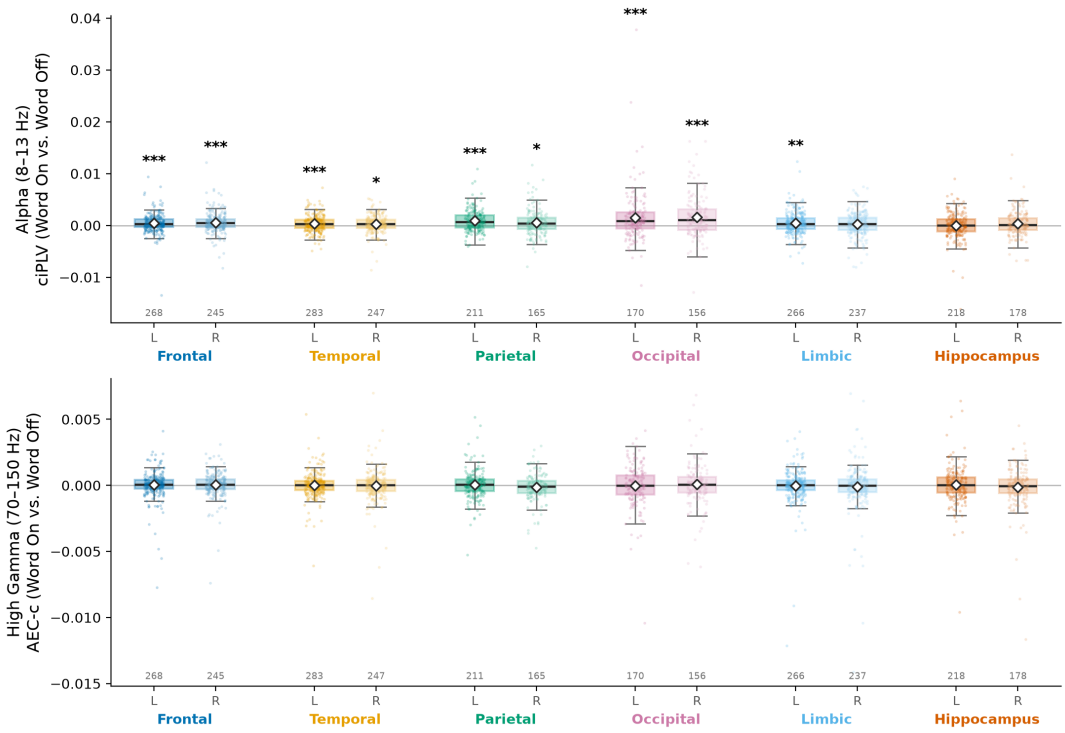


Figure 9. Synchrony change per region, subject distributions.